Benchmarking as a method to evaluate AI systems faces a range of further limitations. First, benchmark datasets can face the same limitations as any other dataset and be too small, narrow, or biased for the task at hand. Like any other capability layer evaluation, benchmarks are limited in assessing ethical or social risk because they do not capture important context. Benchmarks are limited datasets and often permit narrower inferences than their title implies (Narayanan and Kapoor, 2023; Raji et al., 2021; Schlangen, 2019). This explains why benchmarks may yield incompatible results: one paper describes the "benchmark lottery", whereby model performance may seem high on one benchmark and low on another benchmark purportedly testing the same construct (Dehghani et al., 2021). To avoid narrow benchmarks and running into Goodhart's Law (see Roles and responsibilities), one approach is to create "dynamic", continuously evolving benchmarks, where the underlying datasets are updated over time (see, for example, GEM, Gehrmann et al. (2021); Kiela et al. (2021a)). An additional, emerging practice uses pre-trained models to evaluate the outputs of other AI systems (Bai et al., 2022; OpenAI, 2023b; Wu et al., 2023). This enables dynamic testing and creates some degree of flexibility in what is evaluated. This approach could operationalise a user-specified definition of harm, supplied via a prompt or small fine-tuning dataset, thus creating a versatile avenue for highly bespoke probing of an AI system. This automatised approach allows for faster iteration on how a harm is operationalised and can be simpler than collecting an entirely new static evaluation dataset for each new definition or operationalisation of harm. However, this approach is limited by its use of the same type of models the evaluation seeks to measure. Without grounding in human evaluation, there is a risk of amplifying harmful behaviours that models exhibit but cannot identify (see Model-driven evaluation may fill gaps).

A second set of challenges is that benchmarks are often flawed or constructed without necessary care and documentation, raising validity concerns (Blodgett et al., 2021; Liao et al., 2021). To overcome the narrowness problem, multiple benchmarks can be aggregated into a "test suite" (e.g. HELM, Liang et al. (2022), BIG Bench, Srivastava et al. (2023), Safetykit, Dinan et al. (2022)). However, more complex benchmarks may still not capture relevant context or aspects of the target construct. Furthermore, collapsing multiple tests into a single result can make it harder to interpret results and can mask issues such as disparate performance for different groups, as scores are aggregated (Burnell et al., 2023).

Benchmarks are also limited in terms of external validity – that is, with respect to how well results generalise to novel instances (de Vries et al., 2020). This problem is particularly pronounced in benchmarks that transpose tests designed for human cognition to AI systems, without checking that underlying assumptions hold (Binz and Schulz, 2023; Bubeck et al., 2023; de Wynter et al., 2023; Mitchell, 2023; Schaeffer et al., 2023; Shiffrin and Mitchell, 2023).

Human annotation can be used to develop automated benchmarks, which simulate human ratings on novel outputs. Overall, there is a push toward automating testing to reduce labour and time cost, and to reduce human exposure to potentially harmful model outputs. However, automated tests built on human annotation introduce additional noise, as they often fail to accurately represent human judgements and are inevitably less accurate (Liu et al., 2017; Novikova et al., 2018; van Miltenburg et al., 2020). This situation can then be further compounded if human annotators themselves use AI systems to assist the data annotation work (Veselovsky et al., 2023). Recent participatory efforts aim to address these limitations by creating avenues for meaningful input from people from different cultural backgrounds and better benchmarks to identify cultural tropes or stereotypes in images (Qadri et al., 2023b).

Finally, benchmark testing alone may leave unknown capabilities and failure modes undetected. While AI developers have the most technical knowledge about how generative AI models work, there is still a lack of understanding of what these models are capable of (also referred to as "capability overhang", Shevlane et al.

(2023)). This makes it particularly difficult to foresee new model abilities – properties that materialise as the complexity of the system increases (Wei et al., 2022) – that may cause downstream risk. These may not be possible to predict simply by extrapolating the capabilities of smaller-scale models (Schaeffer et al., 2023; Wei et al., 2022).

Adversarial testing

Adversarial testing, also referred to as "red teaming", refers to risk identification exercises through adversarial attacks on AI models, infrastructure, development and deployment environments, and deployed AI products/systems. It is a mode of evaluation targeted at finding vulnerabilities in AI systems that can be exploited to get an AI system to output harmful content (Brundage et al., 2020; Casper et al., 2023; Ganguli et al., 2022; Millière, 2022; Wei et al., 2023). These attacks can be conducted in any modality or across modalities. They are particularly suitable for assessing model vulnerabilities or the friction users encounter when seeking to obtain harmful material. For example, targeted attacks based on textual inputs can be used to produce harmful image outputs (Ma et al., 2021; Yu and Rieser, 2023). Adversarial testing can be conducted by humans or it can be automated, such as by leveraging large language models to perform adversarial testing (Perez et al., 2022a; Yang et al., 2023; Zou et al., 2023). Much like automated benchmark development based on human data, successful adversarial testing prompts can be reused to automate future testing.

Open-ended probing is a method for surfacing unexpected risks and risks that arise in response to innocuous outputs. In the context of safety evaluation, such exploratory probing is a form of "red teaming": researchers may centre a given risk area, such as eating disorder content, and then perform exploratory prompting of AI systems to explore biases and patterns in model responses (Center for Countering Digital Hate, 2023a). Exploratory safety evaluation may also include the visualisation of thematic clusters in training data to help surface potential risks of representation harms (KnowYourData, Google REVISE, Wang et al. (2021)). Exploring the provenance of data can indicate unexpected risks – for example, to what extent private or copyrighted data is present in the dataset, data which may in turn be leaked or cause privacy risks (Choi et al., 2023; Dodge et al., 2021; Kreutzer et al., 2022; Wang et al., 2022). Open-ended probing is limited in that it does not assess specific harm areas or test hypotheses about particular AI system capabilities. It is often a helpful first step to identify leads for more directed evaluation.

Adversarial testing more broadly is subject to the same limitations as human testing and benchmarking outlined above. In addition, adversarial testing is limited by the imagination, contextual knowledge, and skill as probers seek to compromise the AI system. Despite adversarial testing, novel failure modes are typically discovered when an AI system is released to the broader public (Wei et al., 2023). Automated red teaming presents a highly scalable approach but so far has yielded lower-quality tests than human annotation. Further assurances on the quality and reliability of such tests are therefore needed (Mozes et al., 2023; Perez et al., 2022b).

A.2.2. Human interaction layer

Behavioural experiments

Pathways by which an AI system may cause harm to the person interacting with it can be assessed through controlled studies leveraging psychology, human–computer interaction, or behavioural economics methodologies (Lee et al., 2023b; Tahaei et al., 2023). These methods isolate variables of interest, often in highly controlled settings. They may target outcomes from interactions or mechanisms by which potential harm may occur – such as the common propensity for people to believe misinformation more readily if it aligns with their views (Lodge and Taber, 2013). Behavioural experiments typically aim for some level of generality, rather than isolating specific use cases or application designs. This lends them to the study of potential impacts through human interaction of "general

purpose" technologies, such as AI systems whose application domain and product design is not yet defined.

Experiments face a realism trade-off whereby highly contrived laboratory experiments give better insight into causal mechanisms but may not extend to real-world use cases. Another challenge is that such experiments are typically limited in the number of participants, and small-scale studies may fail to detect harms with small effect sizes, which may only become visible at scale – requiring either passive monitoring or other forms of larger-scale analysis. Results may be further confounded due to unrepresentative participant pools, which are often white, educated, and from industrialised, rich, and democratic locales (Henrich et al., 2010). Similarly to human annotation, behavioural experiments have among their practical limitations the fact that they are time-, labour-, and cost-intensive, and may expose human participants to potential harm. Such experiments also require expertise in experiment design, user interfaces, and internal review processes suited to the ethical considerations that arise in this kind of research, some of which may be rare in organisations that develop AI systems (Jackman and Kanerva, 2016; Jindal, 2022; Zevenbergen, 2020).

User research

User research and usability testing can evaluate the needs and behaviours of users, as well as the functionality and possible externalities that AI systems may create at the point of use.[14] It can be conducted in contrived environments to isolate specific variables, such as the impact of particular features or the user interface of applications. User studies can also be conducted "in the wild", by observing how people use AI systems in real-world scenarios. User testing may include behavioural experiments but can also include interviews, talk-out-loud studies, or surveys.

Experimental user studies resemble psychology and human–computer interaction testing as they require human participants whose experiences are studied either in controlled laboratory conditions or in real-world contexts. Differently from behavioural experiments, user testing focuses on specific applications in particular contexts. As a result, the findings often do not extend to other AI systems. Experimental user testing methods include A/B tests to isolate potential impacts of an AI system design feature. These methods can be used to assess how people actually use an AI system, whether it works equally well for different user groups, and whether interaction with the AI system presents unexpected safety hazards. User testing can also measure risk trajectories over time – for example, by mapping out the risk of people being exposed to harmful content over the course of "user journeys", i.e. over the course of human interaction with an AI system. Such user journeys can also model trajectories of users with malicious intent (Roy et al., 2023). "Adversarial testing", while primarily used to investigate performance failures of AI systems, can focus on particular use cases and shed further light on the friction that people may encounter when trying to use the AI system to malicious ends.

Passive monitoring in user studies may include observing patterns in participant interaction with AI systems, such as when, why, and how people try to put the AI system to use. Such passive monitoring can usefully be complemented by qualitative methods, such as interviews or encouraging participants to "talk out loud" as they use a given application. These, in turn, can be complemented by quantitative approaches, such as surveys. Such mixed-methods approaches can triangulate multiple sources of data for a more nuanced understanding of how people use AI systems and what potential unintended externalities they may experience.

User testing has historically focused on product development, rather than risk evaluation for assurances and to inform responsible decision-making. Or, where such user testing may occur, it is not commonly publicly disclosed, and there is thus lacking a

[14] Note that these tests are not often subject to the same ethics scrutiny as psychology experiments. This has led to user studies that adversely impacted user well-being (Jouhki et al., 2016).